

Project Design Phase - II

Technology Stack (Architecture & Stack)

Date	01 November 2025
Team ID	NM2025TMID04086
Project Name	Streamlining Ticket Assignment for Efficient Support Operations
Maximum Marks	4 Marks

Technical Architecture:

Streamlined Ticket Assignment and Approval Workflow Automation using ServiceNow Platform

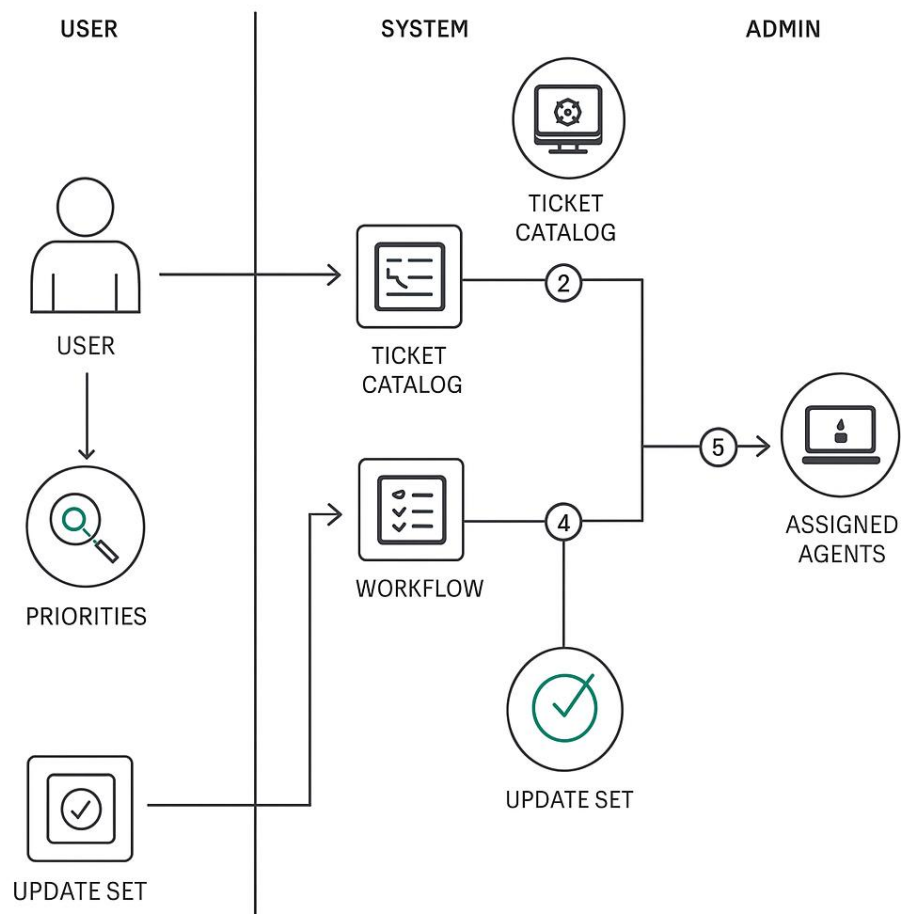


Table-1: Component & Technology:

S.No.	Component	Description	Technology
1.	User Interface	The patient interacts with the system using either Alexa Echo or a mobile device to record daily medication diary responses (e.g., "I took my medicine at 8 PM").	Alexa Echo Device / Mobile Interface
2.	Voice Processing Service	Captures and processes the patient's audio commands through Alexa and routes them to AWS services for interpretation.	Alexa Communications Service
3.	Cloud Contact Center	Handles inbound and outbound voice interactions, transferring patient audio and phone data securely to AWS systems.	Amazon Connect
4.	AI Bot / NLP Engine	Interprets patient speech input (intent, slot, and entities) to understand diary responses such as medication time or type.	Amazon Lex (Intelligent Bot)
5.	Lambda Function (Logic Handler)	Processes recognized data from Lex and stores diary responses in the database. It acts as the middleware logic engine.	AWS Lambda
6.	Database	Stores patient responses, including user ID, medication name, time, and date for reference and analytics.	Amazon DynamoDB
7.	Logging System	Captures interaction transcripts and system events for audit and troubleshooting.	Amazon CloudWatch Logs
8.	File Storage	Stores encrypted audio recordings of patient responses for validation and compliance.	Amazon S3 Bucket
9.	Encryption and Key Management	Secures sensitive patient audio and metadata using managed encryption keys.	AWS Key Management Service (KMS)
10.	Notification Service (Optional)	Sends reminders, alerts, or confirmation messages to patients or caregivers through email or integrated systems.	AWS SNS / Email Integration

Table-2: Application Characteristics:

S.No.	Characteristics	Description	Technology
1.	Open-Source Frameworks	Not applicable, as the solution leverages proprietary AWS cloud and Alexa voice services for automation.	—
2.	Security Implementations	End-to-end encryption of patient audio and data using AWS KMS, secure access control through IAM roles and policies.	AWS KMS, IAM Roles, Secure Access Policies
3.	Scalable Architecture	Built on a cloud-native serverless architecture that scales automatically with user interactions and concurrent requests.	AWS Lambda, Amazon Connect, Amazon DynamoDB
4.	Availability	Ensures continuous availability with fault-tolerant and regionally distributed AWS cloud infrastructure.	AWS Cloud (Multi-Region Availability)
5.	Performance	Delivers fast, real-time responses through optimized Lambda execution, low-latency voice processing, and asynchronous API calls.	AWS Lambda, Amazon Lex, Amazon API Gateway